Artisanal & Small-Scale Gold Mining Business Plan: Traditional vs Modern

Oct 28, 2026 · @BrandWatch

Executive summary

A modern, mercury-free small-scale gold operation recovers roughly twice the gold of a traditional one from the same ore, and earns about 2.3 times the operating profit, for about 14 times the upfront capital. The recommended path is a staged upgrade: start with mercury-free gravity concentration and borax smelting, then add leaching and full water and tailings controls once cash flow and grade are proven.

The plan compares three operating models on the same reference deposit: 10 tonnes of ore per day at 5 g/t gold, 300 operating days a year (3,000 t of ore, 15,000 g of contained gold).

Model	Upfront capital (USD)	Gold recovery	Gold sold per year	Operating profit per year (USD)
Traditional (manual, mercury)	~45,000	30–40%	~6,000 g	~424,000
Stage 1: mercury-free gravity	~160,000	55–65%	~9,000 g	~713,000
Stage 2: modern full plant	~650,000	80–90%	~12,750 g	~977,000

All figures are illustrative planning estimates, in US dollars, at an assumed gold price of USD 3,500/oz (about USD 112.5/g). Replace them with the current spot price, your measured ore grade and local labour, fuel and royalty rates before using this plan with lenders or regulators.

The traditional model looks cheap but carries hidden costs: 50–70% of the gold is lost in tailings, gold is sold at a heavy discount to informal buyers, mercury exposure harms workers and communities, and the operation risks closure or confiscation. The modern model turns the same ore body into a bankable, licensable business that can sell to refineries and certified ethical-gold buyers at near-spot prices.

Business concept, market and legal context

The business mines and processes gold from an alluvial or hard-rock deposit and sells doré bars to a licensed refinery or authorised buyer. Gold has a deep global market, so the business risk lies in production, cost and compliance, not in finding customers.

Business model

- Legal form: a registered company or a formalised cooperative of miners, holding a small-scale or artisanal mining licence for a defined concession.
- Revenue: sale of doré (typically 70–90% gold) priced off the London spot price, less refining charges and buyer margin.
- Secondary revenue: toll-processing ore from neighbouring licensed miners, selling aggregate from waste rock, and certified ethical-gold premiums.

Market and buyers

Buyer channel	Typical price received	Requirements
Informal local trader	75–90% of spot	None, but cash-only and legal risk
Licensed national buyer or central bank programme	90–97% of spot	Licence, production records
International refinery	95–99% of spot	Licence, traceability, OECD due-diligence checks
Certified ethical gold (Fairmined, Fairtrade)	Spot plus a premium (around USD 2,000–4,000/kg, approximate)	Mercury-free or controlled processes, audits, labour standards

Legal and licensing checklist

1. Mining licence or permit for the concession, plus landowner and community agreements.
2. Environmental Impact Assessment (EIA) or Environmental Management Plan, with a reclamation bond where required.
3. Water-use permit and discharge limits.
4. Explosives permits: blasting only by a licensed blaster with licensed storage, or contracted out to a licensed firm.

5. Chemical permits for cyanide purchase, storage and use, aligned with the International Cyanide Management Code.
6. Mercury: the Minamata Convention bans whole-ore amalgamation, open burning of amalgam and cyanide leaching of mercury-contaminated tailings, and requires national plans to reduce and eliminate mercury in mining. Many countries now ban it outright; this plan assumes zero mercury in the modern model.
7. Tax and royalty registration, export permits and gold-trading licences.
8. Occupational health and safety registration, worker contracts and no child labour.

These rules differ by country. Confirm each item with the national mining ministry or regulator before committing capital.

Mining methods: traditional vs modern

Modern methods win on exploration and planning before they win on equipment: knowing where the gold is cuts more cost than any machine.

Exploration and planning (both deposit types)

Stage	Traditional	Modern
Finding gold	Word of mouth, test pits, panning samples	Drone photogrammetry and LiDAR maps, handheld magnetometers, soil geochemistry
Measuring grade	Visual estimate, trial panning	Portable XRF, fire assay at a lab, systematic pit or auger sampling on a grid
Mine plan	Dig where gold was found last	GIS map of grade blocks; mine the richest blocks first, plan waste dumps and backfill
Records	None or a notebook	Daily production log on a phone or tablet, tied to assay results

A. Alluvial (placer) mining

Gold sits in loose river gravels, terraces or surface soils, so the work is moving and washing material.

Element	Traditional	Modern
Excavation	Shovels, hoes, head-pans; deep unprotected pits	Small excavator (5–20 t) or backhoe; benched, sloped pit walls
Washing	Hand panning, hand-fed wooden sluices	Hydraulic sluicing with water monitors, trommel screen to remove oversize, gravel pump
Gold capture	Sacks or blankets on sluices; mercury added to pans	Riffled sluice boxes with modern gold-trapping mats (e.g. vortex or nugget-trap matting), then a centrifugal concentrator on fines
River work	Divers with suction hoses	Small licensed suction dredges with settling barges, only where permitted
Throughput	1–5 m^3 of gravel per day per team	50–300 m^3 per day per wash plant
Land after mining	Open pits filled with stagnant water	Topsoil stockpiled, pits backfilled and replanted as mining moves along

B. Hard-rock (primary) mining

Gold is locked in quartz veins or host rock, so the work is breaking, hoisting and crushing rock.

Element	Traditional	Modern
Access	Narrow hand-dug shafts, often unsupported	Engineered shaft or decline with timber or steel supports and rock bolts
Rock breaking	Hammers, chisels, fire-setting, unlicensed explosives	Pneumatic jackhammers or battery rock drills; controlled blasting only by a licensed blaster
Hoisting	Hand windlass, buckets on ropes	Electric or diesel haulage winch with safety brake; ore cars on rails for drives

Element	Traditional	Modern
Ventilation	None; workers exposed to dust and fumes	Mechanical fans and ducting; wet drilling to keep down silica dust
Selectivity	Large amounts of waste mixed with ore	Follow the vein using grade-control sampling to send less waste to the mill
Throughput	0.5–3 t of ore per day	10–50 t per day

Choosing a method. Alluvial deposits need less capital and pay back faster but run out quicker and disturb more land per gram. Hard-rock deposits last longer and suit a fixed processing plant, but need more capital, more safety controls and a measured resource before investment.

Processing and recovery chain: traditional vs modern

Processing is where most money is won or lost: a traditional circuit leaves 50–70% of the gold in its tailings, while a well-run modern circuit loses 10–20%.

The chain is the same in both models; the tools and the recovery at each step differ.

1. **Size reduction (comminution).** Liberates gold grains from rock.
    - Traditional: hand crushing with hammers, then hammer mills or stamp mills run dry, producing uneven particle size and heavy silica dust.
    - Modern: jaw crusher to about 10 mm, then a ball mill run wet in closed circuit with a classifier or hydrocyclone, grinding to a target size (often 75–150 microns) set by test work. Wet milling also controls dust.
2. **Gravity concentration.** Uses gold's density (about 19 g/cm³ versus about 2.7 for quartz).
    - Traditional: sluices lined with sacks or blankets, then panning; fine gold washes away.
    - Modern: centrifugal concentrator (Knelson, Falcon or iCon type) on the mill discharge, a shaking table to upgrade the concentrate, and jigs for coarser material. Concentrate is a small, high-grade mass that is easy to secure.
3. **Extraction and refining.**

- Traditional: mercury amalgamation of whole ore or concentrate, then burning the amalgam in the open. Recovers only coarse gold, releases mercury vapour and leaves mercury-contaminated tailings.
- Modern, high-grade concentrate: direct smelting with borax flux. No toxic chemicals, low cost, and recovery on clean concentrates is comparable to or better than amalgamation.
- Modern, fine gold in tailings: cyanide vat leaching or carbon-in-leach (CIL) tanks with activated carbon, operated under a licence, in lined vats, with cyanide destruction before tailings disposal. Lifts total recovery to 80–90%.
- Emerging lower-toxicity options: glycine-based and thiosulfate leaching, and halide (chlorine-based) processes. They are promising but need ore-specific test work and are less proven at small scale.

4. **Smelting and off-take.**
   - Traditional: crude melting with charcoal or a blow torch; low-purity sponge gold sold for cash.
   - Modern: induction or gas-fired crucible furnace with borax, soda ash and silica flux, producing weighed, sampled and numbered doré bars sold with assay certificates to a licensed buyer or refinery.

Recovery by step (reference ore, 5 g/t)

Circuit	Gravity recovery	Leach recovery from tailings	Total recovery	Gold recovered per tonne
Traditional (hammer mill, sluice, mercury)	30–40%	None	30–40%	1.5–2.0 g
Stage 1: ball mill, centrifugal concentrator, shaking table, borax smelting	55–65%	None	55–65%	2.75–3.25 g
Stage 2: Stage 1 plus CIL or vat leach on tailings	55–65%	25–30% of feed	80–90%	4.0–4.5 g

Recovery depends strongly on the ore. Before buying equipment, send a 50–100 kg bulk sample for gravity-recoverable-gold and leach tests at a metallurgical lab; this costs a few thousand dollars and avoids buying the wrong plant.

Operational and environmental infrastructure

Infrastructure is what keeps a modern site licensed and running: water, tailings, power and reclamation together take about 40% of Stage 2 capital and pay for themselves through recycled water, lower fuel bills and avoided fines.

Area	Traditional	Modern	Cost-saving angle
Water	Drawn straight from the river; muddy water returned untreated	Storage basin, 3–4 stage settling ponds, closed-loop recirculation (70–90% of water reused)	Less pumping and fewer permit problems in the dry season
Tailings	Dumped on land or in streams, often with mercury	Lined tailings pond or engineered TSF; cyanide destroyed before discharge; dry-stacking where water is scarce	Tailings stay available for re-processing as recovery improves
Power	Small diesel engines on each machine	Solar–diesel hybrid microgrid with batteries, or grid connection	Diesel is often 25–40% of operating cost; solar cuts it by half or more
Reclamation	None; pits abandoned	Concurrent reclamation: topsoil stockpiled, pits backfilled, land replanted as mining advances	Spreads cost over the mine life and protects the licence and reclamation bond
Security	Gold handled by many people	Locked concentrate room, CCTV, sealed concentrate cans, two-person rule	Cuts theft, often a hidden 5–15% loss
Site layout	Unplanned	Mine, plant, ponds, camp and stores laid out to shorten haul and pumping distances	Lower fuel and labour per tonne

Modern-innovation highlights

- Modular, skid-mounted or containerised plants that can be moved to a new deposit.
- Solar hybrid power with remote monitoring of fuel use and generator hours.
- Water-level and turbidity sensors on settling ponds.
- Phytoremediation (planting vetiver and other grasses) to stabilise pit walls and old tailings.
- Digital traceability of each bar, linking mine, date, weight and assay, which buyers increasingly require.

Side-by-side comparison

Traditional mining wins only on entry cost and simplicity; modern mining wins on every measure that decides long-term profit.

Factor	Traditional ASGM	Modern small-scale
Upfront capital	Low (USD 20,000–50,000)	High (USD 150,000–700,000+)
Gold recovery	30–40%	80–90%
Throughput	1–5 t/day	10–50 t/day
Labour	Many low-skilled workers	Fewer, trained operators
Cash cost per gram sold	~USD 28/g	~USD 32/g (~USD 43/g with depreciation)
Price received	75–90% of spot	95–99% of spot, plus possible ethical premium
Access to finance	Almost none	Equipment leasing, bank loans, offtake pre-finance
Health and safety	High risk: mercury, silica dust, shaft collapse	Controlled: wet milling, ventilation, PPE, ground support
Environment	Mercury pollution, turbid rivers, abandoned pits	Mercury-free, recycled water, lined tailings, reclamation
Legal risk	High (closure, confiscation, fines)	Low when licensed and compliant
Mine life	Short; rich parts high-graded and the rest lost	Longer; lower-grade ore becomes economic

Factor	Traditional ASGM	Modern small-scale
Tailings value	Lost or contaminated	Stockpiled clean for future re-treatment

Modern technology map: stage by stage

Modern ASGM technology now covers every stage from exploration to sale, and the biggest gains come from four groups: mercury-free recovery, precision exploration, safer leaching and digital traceability. The tables below list each technology by stage; "Brief" marks those in the technology brief, "Added" marks important ones it left out.

Cost bands are approximate: Low = under USD 5,000, Medium = USD 5,000–50,000, High = over USD 50,000.

Stage 1: Exploration and resource definition

Technology	What it does	Cost band	Source
Drone LiDAR and photogrammetry	Maps terrain, gravel channels and old workings in centimetre detail	Medium	Brief
Drone aeromagnetics	Traces quartz veins, faults and magnetic host rocks before digging	Medium	Brief
Satellite mineral mapping (Sentinel-2, Landsat-9)	Spots hydrothermal alteration zones linked to gold, often with free data	Low	Brief
Portable XRF analyser	Gives on-site element readings (arsenic, copper and other pathfinders) to guide sampling	Medium	Brief
Ground magnetometer and IP/resistivity surveys	Locates sulphide-rich veins and buried structures from the surface	Medium	Added
Ground-penetrating radar	Finds bedrock depth and paystreak traps under alluvial gravels	Medium	Added

Technology	What it does	Cost band	Source
Auger, RC or small diamond drilling	Takes samples at depth to measure tonnes and grade before investing	Medium–High	Added
Soil and stream-sediment geochemistry on a grid	Builds a grade map at very low cost per sample	Low	Added
GIS and 3D resource modelling software	Turns samples into a block model and mine plan	Low	Added
Bulk metallurgical testing (GRG and bottle-roll leach tests)	Shows how much gold gravity and leaching will recover before buying a plant	Low–Medium	Added
AI prospectivity mapping	Combines satellite, geophysics and geology data to rank targets	Low–Medium	Added

Note: handheld XRF is a guide, not a gold assay. It reads gold poorly at typical ore grades, so decisions on grade still need fire assay or PhotonAssay.

Stage 2: Mining and extraction

Technology	What it does	Cost band	Source
Compact excavators and mini-loaders	Mechanise digging and loading in alluvial pits and small open cuts	High (or lease)	Added
Trommel and vibrating screens	Remove clay and oversize boulders before sluicing, raising sluice capture	Medium	Added
Portable wash plants with gravel pumps and water monitors	Process 50–300 m^3 of gravel per day in one mobile unit	Medium–High	Added
Battery-powered and pneumatic rock drills with wet drilling	Faster drilling, less dust, no exhaust underground	Medium	Added

Technology	What it does	Cost band	Source
Non-explosive rock breaking (expansive grout, hydraulic splitters)	Breaks rock where blasting is unsafe or unpermitted	Low–Medium	Added
Electric haulage winches with safety brakes	Safe hoisting of ore and workers in shafts	Medium	Added
Rock bolts, mesh and engineered timber sets	Support shafts and drives against collapse	Low–Medium	Added
Mechanical ventilation fans and ducting	Remove dust, blast fumes and heat underground	Medium	Added
Geo-fenced IoT tracking of machines	Keeps work inside the concession and logs operating hours	Low	Brief

Stage 3: Size reduction (comminution)

Technology	What it does	Cost band	Source
Jaw crushers (portable or skid-mounted)	Reduce rock to under about 10–20 mm	Medium	Added
Ball mills in closed circuit with hydrocyclones	Grind to a controlled size so gold is freed but not over-ground	Medium–High	Added
Wet pan mills and roller mills	Low-cost wet grinding for small plants	Low–Medium	Added
Smart dosing and automated slurry control	Keeps water-to-solids ratio and feed rate steady, preventing sluice blinding	Medium	Brief
Variable-speed drives on mills and pumps	Match motor speed to load, cutting power use	Low–Medium	Added

Stage 4: Gravity concentration

Technology	What it does	Cost band	Source
Centrifugal concentrators (Knelson, Falcon, iCON)	Use high G-force to catch fine free gold without chemicals	Medium	Brief
Modular shaking tables (e.g. Gemini type) with variable-speed drives	Upgrade concentrate to smeltable grade	Low–Medium	Brief
Automated jigs	Recover coarser gold continuously	Low–Medium	Brief
Modern sluice matting (vortex and nugget-trap mats)	Trap fine gold far better than sacks or blankets	Low	Added
Spiral concentrators	Low-cost, no-power pre-concentration of large volumes	Low–Medium	Added
Magnetic separators	Remove magnetite from black-sand concentrates before smelting	Low	Added
Small flotation cells	Recover gold locked in sulphide minerals that gravity misses	Medium	Added
Mercury retorts and fume capture (transition only)	Capture mercury where it is still in use during a phase-out plan	Low	Added

Stage 5: Chemical extraction and leaching

Technology	What it does	Cost band	Source
Glycine and thiosulfate leaching	Lower-toxicity alternatives to cyanide; ore-specific testing needed	Medium	Brief
Modular containerised vat leaching with secondary containment	Mobile, closed-loop leaching of tailings and low-grade ore	Medium–High	Brief
Carbon-in-leach (CIL) and carbon-in-pulp (CIP) tanks	The standard high-recovery method for fine gold	High	Added

Technology	What it does	Cost band	Source
Intensive leach reactors	Leach gravity concentrates quickly in a small, sealed unit, replacing amalgamation	Medium–High	Added
Carbon elution and electrowinning	Strip gold from loaded carbon and plate it as sludge for smelting	Medium	Added
Cyanide destruction (SO2/air or hydrogen peroxide)	Breaks down cyanide before tailings leave the plant	Medium	Added
Halide (chloride-based) leaching	Cyanide-free option for some ores	Medium	Added
Bio-oxidation or ultra-fine grinding	Unlocks gold in refractory sulphide ores	High	Added

Stage 6: Smelting, refining and assay

Technology	What it does	Cost band	Source
Borax direct smelting	Melts gravity concentrate to doré without mercury	Low	Brief
Propane or induction furnaces	Clean, controllable melting; induction gives better recovery and safety	Low–Medium	Added
Fume extraction and scrubbers	Protect smelter operators from flux and metal fumes	Low–Medium	Added
On-site fire assay or PhotonAssay services	Accurate grade for ore, tailings and bars	Medium (or outsourced)	Added
XRF gold purity analysers and density testing	Verify doré purity before sale, strengthening price negotiation	Medium	Added

Stage 7: Power, automation and maintenance

Technology	What it does	Cost band	Source
Solar–diesel hybrid microgrids with battery storage	Cut diesel use and keep the plant running continuously	High	Brief
Smart energy and fuel monitoring	Tracks fuel theft and generator efficiency	Low	Added
Condition monitoring sensors (vibration, temperature)	Predict failures in mills, crushers and pumps before breakdown	Low–Medium	Added
Mine and plant management software on tablets	Logs tonnes, grade, recovery, downtime and stock in one place	Low	Added

Stage 8: Water, tailings and reclamation

Technology	What it does	Cost band	Source
Zero-discharge closed-loop water systems	Recycle process water and stop turbid discharge	Medium	Brief
Flocculants and thickeners	Settle fine solids fast so water can be reused	Low–Medium	Added
Filter presses and dry-stacking	Produce stackable tailings and recover most water	Medium–High	Added
HDPE-lined tailings ponds with leak detection	Contain tailings safely	High	Added
Phytoremediation and assisted re-vegetation	Stabilise and partly clean mined land and old tailings	Low	Added
Water-quality and turbidity sensors	Prove compliance with discharge permits	Low	Added

Stage 9: Health and safety

Technology	What it does	Cost band	Source
Portable gas detectors (oxygen, carbon monoxide, HCN)	Warn of bad air underground and in the leach area	Low	Added
Real-time dust monitors	Measure silica exposure and check controls work	Low–Medium	Added
Mercury vapour analysers	Survey legacy contamination and worker exposure	Medium	Added
Wearable trackers and underground communication	Locate workers and call for help in emergencies	Low–Medium	Added

Stage 10: Traceability, market access and finance

Technology	What it does	Cost band	Source
Blockchain mine-to-market tracking apps	Log origin, GPS, mass and purity at the pithead as a digital passport	Low	Brief
OECD due-diligence and certification systems (Fairmined, Fairtrade)	Unlock refinery access and ethical-gold premiums	Low–Medium	Added
CCTV, biometric access and sealed concentrate containers	Cut gold theft between plant and sale	Low–Medium	Added
Mobile money and digital payroll	Safer payments and auditable labour records	Low	Added
Equipment-as-a-service and gold-linked finance	Spread capital cost through leases, offtake pre-finance or gold-backed loans	n/a (financing)	Added

A note on recovery claims. Gravity tools alone typically recover 40–70% of the gold, depending on the ore. The 80–95% figures in the brief are achievable only when gravity is combined with leaching of the fine gold, as in Stage 2 of this plan. Borax smelting also still needs good ventilation and fume extraction.

Financial plan

On the reference deposit, the modern plant earns about USD 977,000 a year in operating profit versus USD 424,000 for traditional methods, and repays its USD 650,000 capital in about 8 months of full production.

Key assumptions

- Ore: 10 t/day, 5 g/t, 300 days/year = 3,000 t and 15,000 g of contained gold.
- Gold price: USD 3,500/oz (USD 112.5/g), an illustrative planning figure; update to the current spot price.
- Price received: traditional 88% of spot (USD 99/g), Stage 1 95% (USD 107/g), Stage 2 97% (USD 109/g).
- Royalty: 3% traditional (partly informal), 5% for licensed operations; adjust to your country's rate.
- Capital excludes underground or excavation equipment (USD 80,000–150,000), assumed leased or contracted, and excludes 3 months' working capital.

Capital budget (USD)

Item	Traditional	Stage 1	Stage 2
Crushing	8,000 (hammer mill)	25,000	60,000
Grinding (ball mill)	n/a	50,000	90,000
Gravity (sluice, concentrator, table)	4,000	42,000	50,000
Leaching plant (CIL or vat, carbon, elution)	—	—	150,000
Smelting furnace	1,000	6,000	20,000
Power	12,000 (small diesels)	20,000 (generator)	120,000 (solar hybrid)
Water and settling ponds	2,000	10,000	40,000
Tailings storage (lined)	—	—	80,000
Mining tools, PPE, safety	18,000	7,000	—

Item	Traditional	Stage 1	Stage 2
Assay lab	—	—	25,000
Contingency	—	—	15,000
Total	**~45,000**	**~160,000**	**~650,000**

Annual operating budget and profit (USD)

Line	Traditional	Stage 1	Stage 2
Gold sold (g)	6,000	9,000	12,750
Revenue	594,000	963,000	1,389,750
Labour	60,000	79,200	108,000
Fuel and energy	45,000	50,000	60,000
Reagents and consumables (mercury / borax, media / cyanide, carbon, lime)	12,000	18,000	45,000
Maintenance and spares	20,000	25,000	45,000
Water, tailings and reclamation	—	—	25,000
Royalty and levies	17,820	48,150	69,488
Assay, compliance, insurance	—	15,000	30,000
Admin and other	15,000	15,000	30,000
Total operating cost	**169,820**	**250,350**	**412,488**
Operating profit (before tax and depreciation)	**424,180**	**712,650**	**977,262**
Cash cost per gram sold	28.3	27.8	32.4
Break-even ore grade (cash)	~1.4 g/t	~1.3 g/t	~1.5 g/t
Simple payback on capital	~1–2 months	~3 months	~8 months

Add 3–6 months of construction and commissioning before payback starts, and income tax at your local rate. Stage 2 depreciation over 5 years (USD 130,000/year) raises its full cost to about USD 43/g.

Sensitivity of annual operating profit (USD)

Scenario	Traditional	Stage 2
Base case (USD 3,500/oz, 5 g/t)	424,000	977,000
Gold price −20% (USD 2,800/oz)	309,000	713,000
Gold price +20% (USD 4,200/oz)	539,000	1,241,000
Grade falls to 3 g/t	194,000	449,000

Grade is the biggest risk. A 40% drop in grade cuts profit by more than half in both models, which is why measuring the resource before buying a plant matters more than any equipment choice.

Funding options

- Owner equity and cooperative member contributions for Stage 1.
- Equipment leasing or vendor finance for mills, concentrators and solar systems.
- Offtake pre-finance from a refinery or gold buyer, repaid in gold.
- Development-finance and responsible-gold programmes (for example planetGOLD country projects) that support mercury-free conversion.

How to minimise cost

The largest savings come from not spending on the wrong ore or the wrong plant, then from energy, mining selectivity and maintenance.

Strategy	How	Typical saving
Test before you buy	Grid sampling, assays and a metallurgical bulk test before ordering equipment	Avoids six-figure mistakes in plant size or process
Stage the investment	Start with Stage 1 gravity and borax; add leaching only when grade and cash flow justify it	Cuts upfront capital by about 75%
Buy used or refurbished	Reconditioned crushers, mills and generators from closing mines or dealers	30–50% off new prices

Strategy	How	Typical saving
Lease, don't buy, mobile fleet	Rent excavators and compressors by the hour or month	Converts capital to variable cost; no idle-machine cost
Switch to solar hybrid power	Solar plus batteries for daytime milling, diesel as backup	40–60% lower fuel spend; fuel is often a quarter or more of costs
Mine selectively	Follow veins with grade control; screen out barren gravel before washing	Less waste hauled, crushed and ground per gram
Recycle water	Closed-loop recirculation from settling ponds	Lower pumping cost and no dry-season shutdowns
Preventive maintenance	Daily checklists, scheduled liner and bearing changes, critical spares on site	Fewer breakdowns; downtime is the costliest hidden expense
Bulk and group purchasing	Cooperatives buy fuel, grinding balls, borax and PPE together	10–20% on consumables
Train a smaller, skilled crew	Fewer, better-paid operators with clear roles and bonuses	Higher output per worker, lower accident and theft losses
Shared or toll processing	Use a central licensed plant instead of building one, or sell spare capacity	Avoids capital or earns extra revenue
Stay compliant	Permits, safety and environmental controls in place from day one	Avoids fines, closure and confiscation, the most expensive outcome

How to maximise profit

Every 10 percentage points of extra recovery on the reference deposit adds about 1,500 g of gold, worth roughly USD 160,000 a year, at almost no extra cost, so recovery is the first profit lever.

1. **Raise recovery.** Grind to the size test work specifies, run centrifugal concentrators on mill discharge, clean concentrates on a shaking table, and leach fine gold in tailings. Assay tailings weekly; if they carry more than about 0.5 g/t, fix the circuit.

2. **Get a better price.** Sell to licensed buyers or refineries with assay certificates instead of informal traders: moving from 88% to 97% of spot adds about USD 129,000 a year at Stage 2 volumes.
3. **Earn ethical-gold premiums.** Mercury-free, traceable, fairly employed production can qualify for Fairmined or Fairtrade certification and premiums per kilogram.
4. **Re-treat old tailings.** Traditional tailings often still hold 1–3 g/t of gold. Clean, mercury-free stockpiles can be processed at low mining cost; mercury-contaminated tailings need specialist clean-up first.
5. **Increase throughput.** Once recovery is stable, remove bottlenecks (usually the mill or the power supply) to spread fixed costs over more tonnes.
6. **Mine the best ore first, within the plan.** Sequence grade blocks to bring cash forward and repay loans faster, without abandoning lower-grade ore the plant can still process profitably.
7. **Add side revenue.** Toll-process ore for neighbouring licensed miners, sell crushed waste rock as aggregate, and sell surplus solar power to the camp or community.
8. **Control gold security.** Seal concentrates, restrict access to the gold room and reconcile ore tonnes, grade and gold produced daily, so leakage shows up immediately.
9. **Manage price risk.** Sell regularly rather than holding stock, and consider forward sales with your buyer for part of production when prices are high and the loan must be repaid.
10. **Keep the licence.** Community benefit agreements, local hiring and visible reclamation protect the social licence that the whole business depends on.

Health, safety and key risks

The three challenges named for this project — capital intensity, recovery efficiency and occupational health and safety — each have a direct, affordable control.

Risk	Impact	Control
Capital intensity	Equipment fails in harsh conditions; money runs out before production	Stage 1 first; buy robust, locally serviceable brands; keep critical spares; lease mobile equipment; hold 3 months′ working capital

Risk	Impact	Control
Poor recovery	Up to 50% or more of free gold lost to tailings	Metallurgical test work; correct grind size; centrifugal concentrators; weekly tailings assays; operator training
Silica dust (silicosis)	Permanent lung disease	Wet drilling and wet milling, enclosed crushers with water sprays, P2/N95 respirators, ventilation, regular health checks
Ground collapse	Fatalities underground or in pit walls	Engineered shafts with supports, benched pit walls, daily ground inspections, no undercutting
Explosives	Injuries, theft, licence loss	Licensed blaster or contractor only; licensed magazine; strict issue-and-return records
Cyanide	Acute poisoning, environmental spills	Licensed supply, locked storage, pH control, trained staff, antidote kits and emergency plan, cyanide destruction before discharge
Mercury (legacy)	Neurological harm, contaminated land	Zero mercury on site; buy-back and safe disposal of existing stocks; test and isolate old tailings
Grade below plan	Profit falls sharply	Measured resource before major investment; regular reconciliation
Gold price fall	Lower revenue	Low-cost operation; partial forward sales; keep cash reserves
Theft and leakage	5–15% hidden losses	Sealed concentrates, gold room controls, daily reconciliation
Community conflict	Blockades, licence loss	Community agreements, local jobs, grievance process, visible reclamation
Regulatory change	Permit suspension	Full compliance and good regulator relations; monitor legal changes

Minimum PPE per worker: hard hat, safety boots, gloves, eye protection, hearing protection, dust respirator, high-visibility clothing; plus chemical-resistant gear and face shields in the leach and smelting areas. Run a short safety talk every shift and keep an accident and near-miss log.

Implementation roadmap and KPIs

The plan moves from permits to full modern production in about 18–24 months, with each phase gated by a result rather than a date.

1. **Months 0–6: Prepare.** Form the company or cooperative, secure the licence, EIA and community agreements, sample the deposit and run metallurgical tests. *Gate: measured grade and recovery support Stage 1 economics.*
2. **Months 6–9: Build Stage 1.** Install crusher, ball mill, centrifugal concentrator, shaking table, borax furnace, settling ponds; train the crew; remove all mercury from site. *Gate: plant runs at design throughput.*
3. **Months 9–15: Operate and prove.** Run Stage 1, reconcile production daily, build cash reserves, start selling to licensed buyers, stockpile clean tailings. *Gate: six months of stable production and tailings assays confirming leachable gold.*
4. **Months 15–24: Build Stage 2.** Add leach plant, lined TSF, solar hybrid power, water recirculation and assay lab; obtain cyanide permits. *Gate: compliance inspection passed.*
5. **Month 24 onward: Optimise.** Pursue certification, re-treat stockpiled tailings, add toll processing, and reclaim mined-out areas continuously.

Key performance indicators

KPI	Target
Overall gold recovery	Stage 1 ≥ 60%; Stage 2 ≥ 85%
Tailings grade	< 0.5 g/t (Stage 1), < 0.3 g/t (Stage 2)
Plant availability	≥ 90% of scheduled hours
Cash cost per gram sold	Below USD 35/g at reference grade
Price received	≥ 95% of spot
Water recycled	≥ 70%

KPI	Target
Diesel use per tonne milled	Down 40% after solar installation
Lost-time injuries	Zero; all near-misses logged
Mercury on site	Zero
Land reclaimed	Equal to land newly disturbed each year

Review these KPIs monthly with the team and quarterly with lenders, buyers and the community.